Supplementary Material for RelationalRAG-Tables: A Learned Join Verifier for QA over Tables and Knowledge Graphs

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ACML 2026 Journal Track (*Machine Learning*)

1 Scope of this Supplement

We collect supporting experiments for our analysis of matching conditions, including per-seed results and stress tests. All values are measured outputs from the harness or the post-adjudication correction snapshot. Unless marked as multi-seed, results are single-run diagnostics, distinct from the main paper's 20-seed averages. BIRD component results use the corrected extraction. The cross-encoder and SQL generations retain the original extraction, as noted below.

Reading the results. QA denotes question answering and KG a knowledge graph. ICD-10-CM is the International Classification of Diseases, Tenth Revision, Clinical Modification. ICD-9 is its ninth-revision predecessor. OCR denotes optical character recognition, used here to motivate character corruption. BIRD evaluates text-to-SQL (Structured Query Language). HybridQA combines tables and passages. Exact match (EM) compares normalized answers, and execution accuracy (EX) compares SQL query results. Recall@k is the fraction of queries whose gold candidate appears in the top k. Precision is the true-match fraction among accepted pairs, recall the accepted fraction of true matches, and F1 their harmonic mean. AUROC is the area under the receiver operating characteristic curve. Expected calibration error (ECE) is the bin-weighted probability–frequency gap. Coverage is the answered fraction, risk the error rate within that fraction, and AURC the area under the risk–coverage curve. FPR and FNR denote false-positive and false-negative rates. SD denotes standard deviation, CI confidence interval, and n.s. a statistically non-significant comparison. We distinguish these measures because a good pair classifier need not select the right candidate or produce correct QA answers.

2 Baseline Definitions

For reproducibility we collect the exact definition of every baseline compared against the learned verifier. We score the same retrieved candidate pairs with each baseline, changing only the matching function. Unless noted, we accept the highest-scoring candidate.

- **Exact**: the synthetic baseline uses substring matching. The ICD code baseline accepts only when the candidate identifier is byte-identical to the table key. **Canonicalized exact** first lowercases and strips punctuation and whitespace. On our data it equals exact, showing that the hierarchy mismatch survives these formatting changes.
- **Prefix / hierarchy rule**: accept the candidate whose code is the longest prefix of the key (or vice versa), the standard ICD category-to-subcode containment. Near-optimal on hierarchical drift, it collapses when no prefix relation holds (cross-vocabulary or corrupted codes).
- **Fuzzy (edit)**: rank by normalized edit similarity $1 - \mathrm{lev}(a, b)/\max(|a|, |b|)$, where a, b are identifier strings, $|\cdot|$ is string length, and lev counts Levenshtein edits.
- **Jaro–Winkler** (Winkler, 1990): rank by Jaro–Winkler string similarity with default parameters.
- **Character term-frequency–inverse-document-frequency (TF–IDF)** (Sung et al., 2020): cosine similarity over character n-gram (length-n sequence) TF–IDF vectors of the two identifiers.

- **Embedding** ("Embed"): cosine similarity over description embeddings. On the synthetic core these are the reference implementation's deterministic feature-hash embeddings (no pretrained model). On the real-data layer they are a pinned pretrained sentence-transformer whose exact checkpoint and version are recorded in the reproduction guide.
- **Fine-tuned cross-encoder**: a MiniLM cross-encoder (Li et al., 2020) fine-tuned, Ditto-style, to read code and description jointly, trained on the same positive (gold) and hard-negative (near-duplicate identifier) pairs as the verifier (Section 7).
- **Fellegi–Sunter** (Fellegi and Sunter, 1969): the classical probabilistic record-linkage matcher based on a log-likelihood ratio. The full estimator is given in Section 16 and the main paper.
- **No-verifier (NV)**: the planner commits the top-ranked retrieved candidate without the verifier gate, isolating what gating prevents.
- **Oracle**: uses gold join labels to provide an upper bound.
- **Learned verifier** (ours): a multilayer perceptron (MLP) over the explicit identifier-pair feature vector of the main paper, optionally an ensemble of K independently trained members with an uncertainty head.

3 Detailed Results Supporting the Main Paper

Tables 1–5 provide detailed values for the main paper. In the synthetic task breakdown, learned gates recover joins but still struggle to recognize absent entities. In mixed ICD drift and corrected BIRD linking, combining signals improves recall over individual string rules. Cross-vocabulary matching instead favors semantic embeddings. The SQL ablation shows why better matching alone is insufficient: adding value hints can hurt unless the pipeline also identifies relevant schema elements. In Table 1, Single, Join, Multi, and Back denote lookup, linking, traversal, and backtracking. Abst_t measures accuracy on unanswerable questions, while Abst. counts all abstentions. UB denotes the gold-label upper bound. The ICD mixture uses hierarchy, OCR-attempt, and cross-vocabulary mechanisms. In Table 4, each ablation adds the named input to the baseline. Pruning retains selected schema elements, and 7B denotes seven billion parameters.

Table 1: Synthetic EM by verifier and task family (seed 7, 1,392 questions, configured drift 60%).

Verifier	Single	Join	Multi	Back	Abst_t	EM	Abst.
Exact-match	1.00	0.43	0.43	0.43	**1.00**	0.569	636
No verifier	1.00	**1.00**	**1.00**	0.40	0.00	0.860	0
Learned	1.00	0.93	0.93	**0.58**	0.08	0.856	50
Deep ensemble	1.00	0.94	0.94	0.57	0.08	**0.861**	47
Oracle (UB)	1.00	1.00	1.00	1.00	1.00	1.000	36

Table 2: Heterogeneous ICD drift: recall@1 over five seeds, with SD as gray subscripts.

Method	recall@1
Exact code	$0.011_{\pm 0.009}$
Fuzzy (edit)	$0.251_{\pm 0.023}$
Prefix / hierarchy rule	$0.258_{\pm 0.030}$
Embedding	$0.689_{\pm 0.035}$
Learned verifier	$0.752_{\pm 0.035}$

4 ICD-10 + Wikidata: OCR-Corruption Sweep

The main paper states that under attempted data-entry corruption the prefix-hierarchy rule collapses while embedding on the surviving description overtakes it. Table 6 sweeps the probability of calling the corruption routine at a single seed. The audit found that some calls were no-ops. This axis records the configured attempt rate, which can exceed the realized fraction of invalid codes. The reported p values compare the verifier with prefix using paired

Table 3: Alias-corrected BIRD recall@1 over five splits (mean ± SD, 387 test instances each).

Scorer	recall@1 (all)	recall@1 (drift)
Exact match	0.808 ± 0.012	0.000 ± 0.000
Fuzzy match	0.666 ± 0.015	0.167 ± 0.038
Embedding cosine	0.860 ± 0.011	0.806 ± 0.031
Learned verifier	$\mathbf{0.939 \pm 0.010}$	$\mathbf{0.905 \pm 0.051}$

Table 4: BIRD development-set EX with a local 7B coder, using the original extractor.

Pipeline	EX	EX (drift)
Ablation (n=800*, 72 drift)*		
baseline (full schema)	0.458	0.292
+ column values	0.476	0.333
+ value hints (all)	0.446	0.347
+ value hints (selective)	0.456	0.319
+ schema pruning	0.479	0.319
+ pruning + values	**0.483**	**0.361**
Full dev (n=1534*, 131 drift)*		
baseline	0.454	0.328
+ pruning + values	**0.477**	**0.420**

Table 5: Cross-vocabulary recall@1: ICD-9 records to the ICD-10 KG (129 held-out records).

Method	recall@1
Exact code	0.008
Fuzzy code	0.008
Prefix rule	0.008
Embedding	**0.907**
Learned verifier	0.899

bootstrap tests. The observed prefix recall falls $0.729 \rightarrow 0.397$ while embedding is flat. The verifier never significantly beats the prefix rule ($p > 0.45$). We retain this as an implementation stress test and do not use it to estimate subtype-specific performance.

Table 6: ICD drift-subset recall@1 by corruption attempt rate (seed 0, 59–69 drifted test records).

Attempt rate	Exact	Fuzzy	Prefix	Embed	Verifier
0.0 (clean)	0.000	0.644	**0.729**	0.458	0.746
0.5	0.015	0.493	**0.594**	0.493	0.580
1.0 (full)	0.032	0.365	0.397	**0.460**	0.413
Verifier vs. prefix: n.s. throughout (p=0.59, 0.86, 0.45)					

5 Heterogeneous Drift: Per-Seed Detail

The main paper reports aggregate performance on a column generated with hierarchy, OCR-attempt, and cross-vocabulary mechanisms. Table 7 gives the five-seed breakdown. The quantities Δ and p compare our verifier with embedding. The verifier beats the prefix rule in all five seeds ($\Delta \approx +0.49$, $p < 10^{-3}$ throughout) and beats embedding in three of five. Relabeling no-op OCR attempts changes no code, gold node, feature, prediction, or aggregate score. We omit the unsupported pre-adjudication per-subtype breakdown.

Table 7: Heterogeneous ICD drift: per-seed recall@1 and paired-bootstrap comparisons against embedding (10^4 resamples).

Seed	Exact	Fuzzy	Prefix	Embed	Verifier	Δ_{emb}	p
0	0.022	0.256	0.261	0.678	0.700	+0.022	0.257
1	0.006	0.228	0.256	0.733	0.794	+0.061	0.016
2	0.022	0.283	0.311	0.672	0.772	+0.100	<0.001
3	0.000	0.222	0.222	0.639	0.772	+0.133	<0.001
4	0.006	0.267	0.239	0.722	0.722	+0.000	0.532
mean	0.011	0.251	0.258	0.689	**0.752**	+0.063	3/5 sig.

6 Robustness to Description Noise

We drop 50% of description words to test dependence on text (Table 8). Verifier drift recall@1 falls from 0.746 to 0.712, and embedding recall from 0.458 to 0.424. The description-free prefix rule is unchanged. We omit the pre-adjudication hierarchy-to-OCR transfer result because no-op edits contaminated its subtype partition.

Table 8: Drift-subset recall@1 with 50% description word-dropout (seed 0).

Method	clean desc.	noisy (50% drop)
Prefix / hierarchy rule	0.729	0.729
Embedding	0.458	0.424
Learned verifier	**0.746**	**0.712**

7 A Stronger Matcher: Fine-Tuned Cross-Encoder

Greater matching capacity might change the comparison with rules. To test this possibility, we replace the feature MLP with a fine-tuned MiniLM cross-encoder that reads code and description jointly (Table 9). On the original BIRD extraction it reaches 0.95 overall and on drift, but it trails lexical matching on HybridQA ($0.205 < 0.235$) and ties the ICD prefix rule ($p \approx 0.89$). Capacity alone thus appears insufficient to resolve the differences between

matching tasks. Because this training run predates the BIRD alias correction, we retain it as an exploratory capacity check. The corrected feature-verifier experiment in the main paper supports our BIRD claim. In Table 9, n/a denotes an unavailable comparison.

Table 9: Fine-tuned MiniLM cross-encoder recall@1. BIRD uses the original extraction.

Task (held-out)	Baseline	Cross-encoder
BIRD value linking, all (387)	n/a	**0.954**
BIRD value linking, drift (38)	n/a	**0.947**
HybridQA top-1 (395)	0.235 (lexical)	0.205

8 HybridQA: Full Retrieval and Reranking

With gold cell-to-passage links and no identifier drift, HybridQA tests our ability to locate an answer-bearing passage. Lexical overlap outperforms dense `bge-small` retrieval at every cutoff in Table 10, reaching recall@20 0.826 versus 0.647. Our learned evidence verifier improves pair discrimination (AUROC 0.610 versus 0.505 and 0.493 for dense and lexical signals), yet lowers top-1 recall to 0.091 versus lexical's 0.249. Distinguishing pairs on average and choosing the best passage from a candidate pool are different demands, and lexical ranking performs better on the latter here.

Table 10: HybridQA retrieval recall@k on approximately 397 development questions.

Retrieval	@1	@5	@10	@20
Dense (`bge-small`)	0.129	0.390	0.511	0.647
Lexical	**0.244**	**0.544**	**0.670**	**0.826**
Hybrid	0.234	0.474	0.647	0.778

9 BIRD Value Linking: Discrimination and Selective Prediction

We evaluate both candidate ranking and pair classification on alias-corrected BIRD. Across five seeds it reaches recall@1 0.939 ± 0.010 overall and 0.905 ± 0.051 on drift. On the held-out seed-4 split (387 test, 44 drifted), recall@1 is 0.951 overall and 0.955 on drift, AUROC is 0.995, and selective prediction raises accuracy on the answered fraction from 0.994 at full coverage to 1.000 at 50% (Table 11).

Table 11: Alias-corrected BIRD pair discrimination and selective accuracy (held-out seed 4).

AUROC	verifier	exact	fuzzy	embed-cos	
	0.995	0.938	0.909	0.829	
Coverage	1.00	0.90	0.80	0.70	0.50
Accuracy	0.994	0.999	1.000	1.000	**1.000**

10 End-to-End Execution Accuracy: Scale and Full Dev

We test the schema-pruning-plus-values pipeline on the full development set and a larger model (Table 12). These runs use the original extractor and evaluate the combined pipeline. In the table, n counts questions, +p+v denotes pruning plus column values, and Δ is the percentage-point gain over baseline. Full-dev and larger-model runs use different question sets. At full dev (n=1534) the schema-pruning-plus-values pipeline lifts EX $0.454 \rightarrow 0.477$ (+2.3), with the drift subset gaining +9.2 ($0.328 \rightarrow 0.420$). A 14B coder improves identically in relative terms (+2.3 EX), showing the same gain at this second model size. We use these runs to measure the relative pipeline effect and diagnose schema-linking errors.

Table 12: BIRD EX with local coders: effects of schema pruning and column values.

Setting (n)	EX base	EX +p+v	Δ	Δ drift
7B, full dev (1534)	0.454	**0.477**	+2.3	+9.2
14B (400)	0.490	**0.513**	+2.3	+6.7

11 Calibration, System-Level Selective Prediction, and Retrieval Noise

We report calibration error, selective prediction on whole QA answers, and the effect of gating when retrieval can omit the gold candidate.

End-to-end selective prediction. Beyond pairwise joins, we rank each answered QA question by the uncertainty of its committed join and defer the most-uncertain tail (Table 13). System-level exact match rises from 0.889 at full coverage to 1.000 at 50% coverage. The curve is noisier than the pairwise one, because entropy is a weak error predictor on the inherently ambiguous joins and because deterministic single-table lookups occupy the high-confidence region.

Table 13: System-level selective prediction: EM on retained QA answers (1,345 initially answered questions).

Coverage	1.00	0.90	0.80	0.70	0.50
System EM	0.889	0.890	0.876	0.913	**1.000**

Retrieval noise. The synthetic study places the gold node in the candidate pool, which inflates the no-verifier baseline. We stress this on bad-join-backtrack questions, where retrieval surfaces a plausible wrong neighbor, by dropping the gold candidate with probability p (Table 14). The no-verifier planner commits the wrong neighbor and returns a silently wrong answer for 0.60 of questions at every p, and it never abstains. The verifier holds the wrong-answer rate lower and converts the growing pool of unrecoverable joins into explicit abstentions, whose rate climbs from 0.02 at p=0 to 0.29 at p=1. In Table 14, NV denotes no verifier and Ens the ensemble. The p=0 row agrees with the backtrack column of the main paper (no-verifier EM 0.40, ensemble 0.57).

Table 14: Retrieval noise on 264 bad-join-backtrack questions. Here p is the gold-candidate removal probability.

	No-verifier			Ensemble		
p	EM	wrong	abst	EM	wrong	abst
0.00	0.40	0.60	0.00	**0.57**	0.41	0.02
0.25	0.40	0.60	0.00	0.51	0.42	0.06
0.50	0.40	0.60	0.00	0.43	0.44	0.13
0.75	0.40	0.60	0.00	0.34	0.45	0.20
1.00	0.40	0.60	0.00	0.25	0.46	**0.29**

Faithfulness. On the same QA run, answers are *faithful* (grounded in retrieved evidence or a deterministic traversal) for 0.94 of non-abstained questions. The deterministic planner returns stored or computed values. The residual faithfulness gap comes from retrieval coverage on single-table lookups.

Cost-aware utility. We score a correct answer +1, a wrong answer $-\lambda$, and abstention 0 to find the coverage that maximizes utility (Table 15). At λ=1, the optimum answers everything. At λ=5, a wrong answer costs five times the reward for a correct answer, and we obtain the highest utility by deferring the most-uncertain 40%. Utility then rises from 0.33 at full coverage to 0.60 at coverage 0.6.

Table 15: Cost-aware utility on synthetic QA as the wrong-answer penalty λ varies.

Wrong-answer penalty λ	Best coverage	Utility (best / full)
1	1.0	0.78 / 0.78
2	1.0	0.67 / 0.67
5	0.6	0.60 / 0.33

12 Threshold Sensitivity, Compute-Matched Uncertainty, KG Incompleteness, and Efficiency

We next examine practical choices in deployment, from threshold stability and ensemble cost to incomplete graphs and the latency of each pipeline stage.

Threshold sensitivity. Sweeping the acceptance threshold τ_v in Table 16 raises precision and lowers recall monotonically, yet F1 stays between 0.75 and 0.77 across $\tau_v \in [0.4, 0.6]$, placing our fixed τ_v=0.5 in a stable region near the F1 optimum.

Table 16: Held-out verifier discrimination by acceptance threshold τ_v.

τ_v	0.3	0.4	0.5	0.6	0.7
Precision	0.657	0.696	0.767	0.808	0.883
Recall	0.923	0.869	0.738	0.702	0.500
F1	0.768	**0.773**	0.752	0.751	0.639

Ensemble cost and calibration. We compare our five-member ensemble with two single-model signals at one-fifth the compute, the entropy of one member and its score-margin heuristic (Table 17). The two signals rank cases identically. A single model matches ensemble accuracy at 50% coverage (0.86) and is slightly higher at full coverage in this diagnostic. The ensemble improves calibration, with ECE 0.027 versus 0.036. These results agree with the saturating K-sweep in the main paper and support a single model when only ranking is needed.

Table 17: Selective accuracy: a five-member ensemble (5× compute) versus a single-model signal (1×).

Coverage	1.00	0.90	0.80	0.70	0.50
Ensemble (5×)	0.725	0.757	0.783	0.803	0.856
Single model (1×)	0.743	0.755	0.779	0.793	0.861

KG incompleteness. Removing a fraction of graph edges (Table 18) lowers exact match as expected, and most lost answers become abstentions. At 10% and 20% edge removal the wrong-answer rate stays at 0.02 and 0.04 while abstention rises to 0.13 and 0.41. Our planner mainly responds by refusing, although wrong answers also increase slightly.

Table 18: KG incompleteness on 792 join and multi-hop questions.

Edges removed	0%	10%	20%	30%
Exact match	0.939	0.845	0.552	0.518
Wrong	0.010	0.023	0.035	0.046
Abstain	0.051	0.133	0.413	0.437

Efficiency. Table 19 breaks per-question latency into stages over 1,392 questions on a 6,509-entry index. Retrieval dominates (50th/95th latency percentiles, P50/P95 10.7/11.6 ms), and execution, which includes the verifier and ensemble scoring, has sub-millisecond median latency (P50/P95 0.6/1.2 ms).

Table 19: Stage-wise CPU latency: 1,392 questions, 6,509 index entries.

Stage	P50 (ms)	P95 (ms)
Retrieval	10.7	11.6
Execution (incl. verifier)	0.6	1.2
Total	11.3	15.2

Where gating hurts. On clean join questions, the gold candidate is present and the ungated planner achieves EM 1.00. Here we see the cost of gating, as our verifier rejects 16 of 264 correct joins near the decision boundary, lowering EM to 0.94, a 6% cost. Raising τ_v increases these rejections while screening more bad joins. HybridQA presents a separate cost, since our learned matching signal also underperforms lexical ranking.

13 Verifier Feature Ablation

If a single cue such as numeric-suffix agreement explained the verifier's performance, retraining on that feature alone should recover it. We test this with single-feature subsets. On synthetic graded corruption (Table 20), digit agreement reaches AUROC 0.70, edit-ratio 0.76, prefix overlap 0.67, and embedding only 0.52. The full combination reaches 0.82, showing a benefit from combining cues even though the generator preserves numeric keys. On real ICD-10 hierarchies (Table 21), prefix and code-structure features already reach recall@1 0.826 overall and 0.729 on drift. Because hierarchy drift is by construction a prefix relation, adding embedding yields only a small further drift gain (full 0.746).

Table 20: Synthetic held-out discrimination after retraining on feature subsets.

Feature subset	Accuracy	AUROC
Digit agreement	0.678	0.699
Edit ratio	0.683	0.757
Prefix overlap	0.637	0.669
Embedding	0.566	0.517
Full	**0.745**	**0.820**

Table 21: ICD-10 recall@1 after retraining on feature subsets.

Feature subset	recall@1 (all)	recall@1 (drift)
Prefix overlap	0.826	0.729
Edit ratio	0.805	0.644
Embedding	0.754	0.458
Code structure	0.826	0.729
Full	**0.826**	**0.746**

14 Verifier Failure Analysis

Table 22 distinguishes pair-label errors from top-ranked retrieval errors. On corrected BIRD pairs (seed 0), FPR is 0.46% and FNR 5.43%. On 236 ICD records, 41 top-ranked candidates are wrong. Of these, 18 confuse sibling codes sharing a three-character category, such as `R508` and `R509`. Our gate commits 38 of all 41 incorrect candidates and rejects 18% of gold pairs. These are distinct counts: sibling ambiguity explains many ranking errors, while high scores on wrong candidates expose overconfidence.

15 Scalability of the Multi-Index Retriever

We scale the pure-Python linear-scan retriever beyond the corpus in Table 19. Its P50 latency rises from 11 ms at 6.5k entries to 206 ms at approximately 104k entries, a 19× increase for a 16× corpus. The verifier's per-pair

Table 22: ICD ranking errors and alias-corrected BIRD pair-classification errors.

ICD-10 join (236 test records, recall@1)	
recall@1 misses	41 (17%)
sibling-code confusion	18
committed a wrong join	38
safely abstained	3
gold pair rejected (FN)	42 (18%)
BIRD value linking (held-out pairs, τ_v=0.5)	
false negatives (gold rejected)	21 (5.43%)
false positives (wrong accepted)	75 (0.46%)

cost stays constant. Retrieval is the scaling bottleneck, motivating approximate nearest-neighbor search and index sharding around 10^5 entries.

16 Record-Linkage Baseline, Composite Keys, Structural Context, and Risk-Controlled Selective Prediction

The record-linkage baseline, composite-key, and risk-controlled selective-prediction studies run on the deterministic synthetic core. The structural-context study runs on the real ICD-10 layer, where the code hierarchy supplies one-hop context.

Classical record-linkage baseline. We benchmark a Fellegi–Sunter probabilistic matcher (Fellegi and Sunter, 1969), the canonical non-neural record-linkage model, on the same features and pairs as the learned verifier. We binarize each join feature into a field-agreement indicator, estimate the agreement probability under a match (m_j) and a non-match (u_j) for each field j from the labelled training pairs with Laplace smoothing, and score a pair by the summed log-likelihood ratio

$$s_{\mathrm{FS}} = \sum_j \mathbb{1}[a_j] \log \frac{m_j}{u_j} + \mathbb{1}[\neg a_j] \log \frac{1-m_j}{1-u_j},$$

where a_j denotes agreement on field j, m_j and u_j are its match and non-match agreement probabilities, and $\mathbb{1}$ is an indicator. Fellegi–Sunter reaches AUROC 0.753, and our MLP on the same features improves AUROC by 0.067. The MLP can combine continuous feature values and their interactions, while this baseline binarizes each field's agreement. Its strongest agreement fields are containment, full-digit match, and digit-suffix match, consistent with the numeric keys our generator preserves.

Composite (multi-attribute) keys: AND/OR composition and conflicts. We combine per-attribute match scores s_1, s_2 to evaluate two-attribute keys (Table 23). AND uses their product with threshold τ_v^2. OR uses $1-(1-s_1)(1-s_2)$ with threshold τ_v, the main paper's acceptance threshold. Positives match on both attributes, conflicts on only one, and negatives on neither. Recall$^+$ is positive acceptance, and conflict accept measures acceptance when only one attribute matches. AND ranks composite matches well (AUROC 0.934 versus 0.737 for a single attribute) and rarely accepts conflicts (0.018), but its positive recall is only 0.428. OR favors recall (0.910) while accepting 0.600 of conflicts. Exact composite matching recovers only 0.3% of positives. These outcomes reflect different key semantics: AND requires both attributes, whereas OR allows either to identify the record. We have not calibrated the combined score under either composition.

Table 23: Two-attribute joins under drift: positive recall and acceptance of conflicting keys.

Method	AUROC	Recall$^+$	Conflict accept
Exact composite	n/a	0.003	0.000
Single-attribute verifier	0.737	n/a	n/a
Per-attribute OR	0.886	0.910	0.600
Per-attribute AND (product)	**0.934**	0.428	0.018

Risk-controlled selective prediction. We compare calibration-based threshold selection for a single member and an ensemble (Geifman and El-Yaniv, 2017; Angelopoulos and Bates, 2023). For target risk α_{risk}, we select the largest set whose calibration-split error meets the target, then evaluate a disjoint test split (Table 24). At target 0.01, ensemble risk is 0.007 versus 0.013 for the single member. At target 0.10, the ensemble covers more cases (0.424 versus 0.399), but both exceed the risk target (0.113 and 0.110). Neither method performs best at every target, and the selected threshold can miss its target on test data. Achieving observed risk below 5% here requires accepting only about one quarter of pairs. The main paper reports the broader 20-seed evaluation.

Table 24: Calibration-selected coverage and test risk: single member versus ensemble at target α_{risk}.

	Single member		Ensemble	
Target α_{risk}	Cov.	Risk	Cov.	Risk
0.01	0.212	0.013	0.197	0.007
0.02	0.212	0.013	0.197	0.007
0.05	0.244	0.044	0.236	0.034
0.10	0.399	0.110	0.424	0.113

Lightweight structural context. We add four ICD hierarchy features to our six-feature verifier: shared three-character category (one-hop parent), shared chapter letter, code-length agreement, and a sibling indicator. Training pairs and ensemble seeds remain fixed. Table 25 shows improved AUROC (0.834 to 0.855) and ECE (0.064 to 0.039), with feature extraction remaining near 6 μs per pair. Yet top-1 recall rises only 0.4 points and sibling confusions decrease by one. We attribute this limited retrieval gain to the shared context. Sibling nodes have the same parent category, allowing this feature to improve probability estimates without resolving which sibling is correct.

Table 25: Adding ICD hierarchy features on a controlled split: discrimination, calibration, recall, errors, and latency.

Verifier	AUROC	ECE	R@1 (all/drift)	Sibling errors	μs/pair
Base (6 features)	0.834	0.064	0.822/0.746	19	6.2
+ structural (10)	**0.855**	**0.039**	0.826/0.746	18	6.0

17 Normalization Baselines, Recalibration, Input-Type Detection, and Robustness

Practical normalization and fuzzy baselines. We compare practical code normalization and fuzzy matching in the consolidated ICD results. Canonicalization strips dots and spaces before exact matching, yielding the same recall as plain exact matching (0.644 all, 0.000 drift). Jaro–Winkler (Winkler, 1990) on canonicalized codes ties the prefix rule (0.826 all, 0.729 drift). Character TF-IDF cosine trails embedding (0.729 all, 0.407 drift), while our verifier reaches 0.746 on drift.

Single-model recalibration. Recalibrating a single model would require only one-fifth the ensemble's compute. We test this alternative with temperature scaling, which fits a scale for the logits, and isotonic regression, which fits a monotone probability map (Table 26). Fitting temperature on a held-out calibration half gives test ECE 0.039, with a fitted temperature near 1. Isotonic regression gives 0.043, consistent with overfitting the calibration half. Both exceed the ensemble's ECE 0.024 on the same split.

Table 26: Test-half ECE after single-model recalibration, compared with a deep ensemble.

Method	ECE
Single, raw	0.039
Single, temperature scaled (T=1.1)	0.039
Single, isotonic	0.043
Deep ensemble	**0.024**

Threshold stability on real data. We repeat the threshold sweep on real ICD pairs (Table 27). As τ_v rises from 0.3 to 0.7, precision rises and recall falls monotonically. F1 stays within $[0.79, 0.80]$ for $\tau_v \in [0.4, 0.6]$, placing our fixed threshold in a stable region, as in the synthetic sweep.

Table 27: ICD pair precision, recall, and F1 by acceptance threshold τ_v.

τ_v	0.3	0.4	0.5	0.6	0.7
Precision	0.732	0.745	0.759	0.829	0.835
Recall	0.881	0.856	0.839	0.758	0.708
F1	**0.800**	0.797	0.797	0.792	0.766

Input-type detector. To distinguish identifier queries from natural-language spans, we train a logistic classifier on token count, digit and alphabetic fractions, mean token length, code punctuation, and a single-token flag. With ICD codes as identifiers and their descriptions as spans, it reaches 0.997 five-fold cross-validated accuracy versus a 0.5 majority baseline. Despite this routing accuracy, the cascade results point to a separate challenge in calibrating match scores across both kinds of input.

Split protocol and leakage. We split ICD training and test sets by record and quantify the remaining overlap. Only 1.3% of test records share their exact billable code with a training record and 19.5% share their gold Wikidata node, while 41.1% share the coarse three-character category. Our verifier reads relational features of each record–node pair without node identity. This prevents direct lookup of training-node labels, although shared categories remain part of the evaluation's structural overlap.

Real-data verifier training. The construction of real-data verifier pairs governs both leakage and generalization. For each training record, we form one positive pair, (record, gold node), and eight negatives by *hard-negative mining*: the four KG nodes whose label-plus-code embedding is closest to the record (the distractors a retriever would actually surface) and four uniformly random nodes. Each ensemble member trains on a bootstrap resample of these pairs with a distinct hidden width, and we use the original binary labels because the ICD pairing is exact at construction (a record's gold node is its true category). The deliberate label ambiguity is confined to the synthetic graded-corruption split, where the same-entity probability decreases with edit distance through a $\sim$50/50 boundary at two edits.

Retrieval noise on real data. The retrieval-noise probe of the controlled study repeats on the real ICD corpus (Table 28). We drop the gold KG node from the candidate pool with probability p and compare the no-verifier planner, which commits the top-scored node, against the verifier, which commits only when the top score clears τ_v. As p rises the no-verifier wrong-answer rate climbs to 1.00, because every dropped-gold record yields a silent wrong join, whereas the verifier holds its wrong rate near 0.25 and converts the rest into abstentions, reaching 0.75 at p=1. The value of gating again grows as retrieval degrades, now on real identifiers.

Table 28: ICD retrieval noise: wrong-answer and abstention rates as gold-node removal probability p increases.

p	0.00	0.25	0.50	0.75	1.00
No-verifier wrong	0.18	0.39	0.59	0.79	1.00
Verifier wrong	0.18	0.18	0.21	0.24	0.25
Verifier abstain	0.00	0.21	0.39	0.56	**0.75**

Enterprise-scale retrieval. We measure retrieval latency over N unit-length 384-dimensional vectors using the FAISS vector-search library (Table 29). We compare an exact flat index with an approximate inverted-file (IVF) index. Exact search grows from 0.04 ms at 10^4 to 28 ms at 10^6 candidates, while approximate search stays below half a millisecond, giving roughly a 60$\times$ gap at 10^6. These are hardware-dependent mean query latencies. The reliable trend is the growth in exact-search cost across index sizes. Our verifier scores only the few candidates returned by the index, keeping its per-join cost constant in N.

Table 29: FAISS query latency: exact versus approximate search over N unit-length 384-dimensional vectors.

N	10^4	10^5	10^6
Exact (flat) ms	0.036	0.344	28.041
Approximate (IVF) ms	0.016	0.012	**0.480**

18 Additional Studies

The main paper reports six further studies whose headline rows appear there. Original outputs are in the artifact (`artifacts/multiseed`, `theory_check.json`, `e1`, `icd_kg/results_e5_cascade.json`, `e6`, `e7`). The reproduction guide identifies the drivers, and the supplied annotation package adds the post-adjudication alias-correction driver and snapshot used for the corrected BIRD rows.

Multi-seed synthetic study (E2). `relational_rag_tables.multiseed` regenerates the corpus and retrains the verifiers for 20 seeds and writes per-seed JSON plus an aggregate with mean, sd, and t-based 95% CI for every scalar, 1%-step risk–coverage curves with 500-resample bootstrap bands, AURC and excess AURC, and validation-half threshold selection. The per-K selective-prediction rows, the retrieval-noise probe, the feature ablation, and KG incompleteness (Tables 17, 14, 20, 18) now live only here.

Theory check (E3). Table 32 reports match prevalence π_+, FPR α, FNR β, and critical penalty $\lambda^\star$. Sign agreement compares predicted and measured utility at $\lambda \in \{1, 2, 5\}$. The table separates the utility threshold from its empirical sign agreement. Small $\lambda^\star$ values do not bound errors after a distribution shift. `relational_rag_tables.theory_check` measures α, β, π_+ on held-out pairs and r, r_0, δ on gold join pairs for every seed and compares the theorems' predictions with the planner's measured utility and join-task EM.

Table 30: Consolidated ICD-10-CM-to-Wikidata baselines: recall@1 overall and on drift.

Method	All ↑	Drift ↑
Exact code match	0.644	0.000
Canonicalized exact	0.644	0.000
Fuzzy (edit) (Winkler, 1990)	0.805	0.644
Jaro–Winkler (Winkler, 1990)	0.826	0.729
Character TF-IDF (Sung et al., 2020)	0.729	0.407
Prefix / hierarchy rule	0.826	0.729
Embedding (Efthymiou et al., 2017)	0.754	0.458
Learned verifier (Mudgal et al., 2018; Li et al., 2020)	**0.826**	**0.746**

Large language model matchers. We keep the core comparison reproducible without API dependence and use sub-millisecond pair verification. Large language model (LLM) calls take 100–500 ms per pair in the reported comparison. The cross-encoder study (Table 9) tests greater matching capacity within a local model.

Additional comparisons and examples. Table 30 consolidates the ICD-10 baselines. For BIRD schema linking with column values (n=800), full-schema EX is 0.476 with execution-error rate 0.131. Embedding-only pruning gives 0.473 (0.141), learned pruning 0.483 (0.131), and oracle gold tables 0.525 (0.114). Individual verifier decisions illustrate both recovery and confusion. We score `ACC-1001`→`ACX1001` at 0.91 for a correct join and `J45.9`→`J45.901` at 0.87 for a hierarchy match. The sibling pair `E11.65` versus `E11.69` receives 0.81 and is incorrectly committed, showing how shared code structure can also produce overconfidence.

Table 31: Heterogeneous ICD drift: fusion recall@1 over five seeds (gray subscripts: SD).

Fusion	recall@1
Tuned cascade prefix→fuzzy→embedding	$0.776_{\pm 0.021}$
Logistic fusion (same features)	$0.771_{\pm 0.022}$
Learned verifier (MLP ensemble)	$0.757_{\pm 0.043}$
Learned verifier + pool-relative features	$0.771_{\pm 0.018}$

Table 32: Measured inputs and utility predictions of the main paper's Theorem 1 and Corollary 1 (held-out pairs, $\tau_v = 0.5$).

Dataset	π_+	α (FPR)	β (FNR)	$\lambda^\star$	sign agrees
Synthetic (20 seeds)	$0.564_{\pm 0.011}$	$0.327_{\pm 0.045}$	$0.206_{\pm 0.038}$	$0.39_{\pm 0.05}$	20/20
BIRD value linking	0.023	0.0046	0.054	0.0013	—

Table 33: Label-noise sensitivity: retrain after corrupting a fraction η of training assignments and evaluate on clean test data ($\tau_v = 0.5$).

η	BIRD r@1 all / drift	BIRD FPR / FNR	ICD r@1 (3 seeds)	ICD FPR / FNR
0.00	0.928 / 0.947	0.46% / 5.4%	$0.752_{\pm 0.042}$	0.09% / 16.7%
0.05	0.920 / 0.895	0.47% / 4.7%	$0.739_{\pm 0.045}$	0.09% / 19.3%
0.10	0.910 / 0.868	0.42% / 5.7%	$0.726_{\pm 0.037}$	0.09% / 22.2%
0.20	0.910 / 0.895	0.26% / 25.6%	$0.733_{\pm 0.035}$	0.05% / 34.1%

Verifier-isolated BIRD ablation (E1). `relational_rag_real.run_e1` ran 16 configurations on a CUDA server with vLLM and an activation-aware weight-quantized (AWQ) 4-bit coder. The configurations comprise the baseline, pruning plus values without hints, five linkers, five drift-only-hint variants, and three stricter thresholds. With drift-only hints, exact matching produces no hints, fuzzy linking lowers EX by 4.4 points at precision 0.01, embedding changes EX by -1.0, and our verifier by -0.3 ($+1.5$ on drift at τ=0.8). Oracle drifted literals yield $+0.5$ EX overall and $+6.1$ on drift. These runs use the submitted extractor. The alias correction changes cross-fit training pools, and the LLM generations were not repeated. EX remains scored against the official gold SQL.

Joint matcher across input types (E4). `relational_rag_real.run_e4_joint` trains one 5-member MLP ensemble on pooled BIRD value-linking and HybridQA evidence pairs with common base features, scores relative to the pool maximum, and input-type features. On the original extraction it improves calibration across tasks and closes two thirds of the HybridQA gap to lexical matching without beating it (Table 34). Because the run predates the BIRD alias correction, we use it as an exploratory routing diagnostic and exclude it from corrected BIRD evidence.

Table 34: E4 joint matcher: recall@1 and pair-level ECE on BIRD and HybridQA over three seeds.

Scorer	BIRD all	BIRD drift	HybridQA	ECE B / H
Lexical (edit ratio / coverage)	$0.710_{\pm 0.023}$	$0.381_{\pm 0.095}$	$0.281_{\pm 0.030}$	—
Dense cosine	$0.832_{\pm 0.025}$	$0.750_{\pm 0.058}$	$0.174_{\pm 0.023}$	—
Containment	$0.925_{\pm 0.009}$	$0.716_{\pm 0.092}$	$0.169_{\pm 0.012}$	—
BIRD-only verifier (24 feats)	$0.924_{\pm 0.006}$	$0.863_{\pm 0.048}$	$0.146_{\pm 0.043}$	0.001 / 0.117
HybridQA-only verifier (24 feats)	$0.343_{\pm 0.090}$	$0.419_{\pm 0.128}$	$0.229_{\pm 0.029}$	0.108 / 0.039
Joint, base + regime feats	$0.934_{\pm 0.006}$	$0.871_{\pm 0.055}$	$0.222_{\pm 0.030}$	0.002 / 0.026
Joint, + pool-relative feats	$0.924_{\pm 0.007}$	$0.863_{\pm 0.048}$	$0.237_{\pm 0.030}$	0.001 / 0.034

Tuned rule cascades (E5). `relational_rag_real.run_e5_cascade` tunes cascade order and thresholds on the training split (prefix, then fuzzy at ≥ 0.9, then embedding in all five seeds). Similar fusion means in Table 31 suggest that complementary signals matter more here than model capacity. Rule-only results appear in Table 2. The aggregate comparison is unchanged by relabeling no-op OCR cases because subtype is not an input to any scorer. We omit the original per-subtype breakdown, which mixed valid hierarchy codes into the OCR stratum.

Label protocol and noise (E6). In Table 33, r@1 is recall@1. Injected training-label errors mainly increase false rejections in this sweep. Low FPR alone does not ensure high recall. The supplied, self-contained `annotation/` package contains the two 100-item samples, completed independent A/B worksheets, offline evidence, provenance, integrity hashes, and standard-library agreement/adjudication scripts. The audit has 100 paired items per dataset: BIRD $\kappa = 0.81$ for protocol-gold correctness (four disagreements, all resolved by the gold SQL) and 1.00 for drift, with adjudicated error rates 0.12/0.00. ICD $\kappa = 1.00$ for drift type and undefined for gold correctness, with adjudicated rates 0.00/0.05. The package also includes the alias-corrected BIRD rerun and its machine-readable output. Table 33 uses the corrected BIRD extraction. We use the injected-noise sweep to measure sensitivity. The main paper propagates the audit corrections separately.

End-to-end table+KG QA (E7). `relational_rag_real.run_e7_icd_qa` builds the records table and the live-fetched Wikidata KG, writes the fixture in the synthetic corpus format (`fixtures/icd_qa/seed_{0..4}`), and runs

the planner with each gate. Per seed: 436 records (21 ghosts), 159 test records, 518 questions (159 lookup, 167 join, 70 backtrack, 101 two-hop, 21 abstain), 2,126 triples. Per-task EM for the verifier gate: join 0.75, backtrack 0.72, two-hop 0.78, abstain 1.00. The corresponding ungated values are 0.90, 0.69, 0.87, 0.00.

19 Reproduction

The reproduction guide names the driver and output file for each table. The annotation package's `recompute_bird_protocol_impact.py` applies the adjudicated alias correction and produces the corrected BIRD component and noise results without modifying the artifact checkout. The remaining real-data and synthetic tables regenerate as documented. Pretrained embeddings and the local coder make real-data runs seeded but not bit-identical across hardware, and reported values are single-seed point estimates except where a mean±sd over seeds is given. E1, E4, and the cross-encoder are explicitly marked as pre-correction exploratory or as-run analyses. They are not used for the corrected BIRD headline.

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